COST OF CAPITAL

Weighted Average Cost of Capital (WACC)

Concept, formula, book value vs market value weights

Learning Goals

- ✓ Explain what the Weighted Average Cost of Capital (WACC) represents and why it matters
- ✓ State and interpret the WACC formula for debt and ordinary equity
- ✓ Extend the formula to firms financed with debt, preferred and ordinary equity
- ✓ Estimate component weights using book values versus market values
- ✓ Calculate WACC for two Indian companies using both weighting approaches
- ✓ Compare the results and explain why market values are the preferred convention

The Cost of Capital

What WACC means and how it is built up

Firm's Cost of Capital

The minimum rate of return the firm must generate to meet the required returns of its investors

Referred to as an opportunity cost of capital

Based on the rates of return that investors earn on investments in firms with a similar risk profile

Calculating the WACC

For a firm financed with debt and ordinary equity, the weighted average cost of capital (WACC) formula is:

$$WACC = r_D(1 - T_C) \times D/(E+D) + r_E \times E/(E+D)$$

r_D	r_E	T_C	D	E
Cost of debt	Cost of (ordinary) equity	Corporate tax rate	Market value of the firm's debt	Market value of the firm's equity

WACC with Debt, Preferred and Ordinary Equity

For a firm financed with debt, preferred equity and ordinary equity, the WACC formula extends to:

$$WACC = r_D(1-T_C)\times D/(E+P+D) + r_P\times P/(E+P+D) + r_E\times E/(E+P+D)$$

r_D	r_P	r_E
Cost of debt	Cost of preferred equity	Cost of ordinary equity

Note: E, P and D should be stated at market value wherever possible.

Estimating the Cost of Debt and Equity

The two building blocks that feed into WACC

Cost of Debt

If a firm has:

Term loan	Issued bonds	Loans & bonds on issue
The lender charges the current interest rate	The bond's yield to maturity — not the coupon rate	A weighted average cost of debt is calculated using the market values of each source

After-tax cost of debt: $r_D (1 - T_C)$

Cost of Ordinary Equity — CAPM

The Capital Asset Pricing Model (CAPM) is the most widely used approach to estimate the cost of ordinary equity:

$$r_E = R_f + \beta_E \times (R_m - R_f)$$

R_f

Risk-free rate — yield to maturity on long-term government (G-Sec) bonds

β_E

Beta of the stock — its systematic risk relative to the market

$(R_m - R_f)$

Market risk premium — long-term average excess return of equities over bonds

Estimating the Weights

Book value vs market value of debt and equity

Target Capital Structure Weights

The most appropriate weights are the target capital structure weights, stated in market value terms.

Target capital structure weights are preferred because:

The firm's current capital structure might not reflect its future (target) capital structure — so market values of debt and equity today are used as the best available proxy for those target weights.

Calculating Debt & Equity Weights

Book value weights

Many managers rely on balance-sheet (book) values of debt and equity, despite the availability of market values — mainly because book values are simple to obtain directly from the financial statements.

Market value weights

Market value of equity = share price × shares outstanding. The book value of debt is commonly used as a proxy for its market value, since the difference between the two is typically small.

Why it matters: Book equity reflects historical accounting values, while market equity reflects investors' current expectations of future performance. For growing or asset-light firms the two can differ enormously — which changes the calculated weights, and therefore the WACC itself.

WACC in Practice

Two Indian company case studies

Company Snapshot — Reliance Industries Ltd

NSE: RELIANCE • Conglomerate — Energy, Retail, Telecom

Beta (β) **1.05**	Risk-free rate (Rf) **7.0%**	Market risk premium **6.0%**
Pre-tax cost of debt **7.5%**	Corporate tax rate **25%**	Market value — Equity (E) **₹18,00,000 Cr**
Market value — Debt (D) **₹3,10,000 Cr**	Book value — Equity (E) **₹8,20,000 Cr**	Book value — Debt (D) **₹3,30,000 Cr**

Figures are illustrative, rounded approximations for teaching purposes based on recent public financial data — not investment advice.

Reliance Industries — WACC Using Market Weights

Component	Cost	Weight	Weighted
Cost of equity: rE = 7.0% + 1.05 × 6.0%	13.3%	85.3%	11.35%
After-tax cost of debt: rD(1−TC) = 7.5% × 0.75	5.6%	14.7%	0.82%

Market Value WACC = 12.2%

Weights: E/(E+D) = 18,00,000/21,10,000 = 85.3% D/(E+D) = 3,10,000/21,10,000 = 14.7% (₹ Crore)

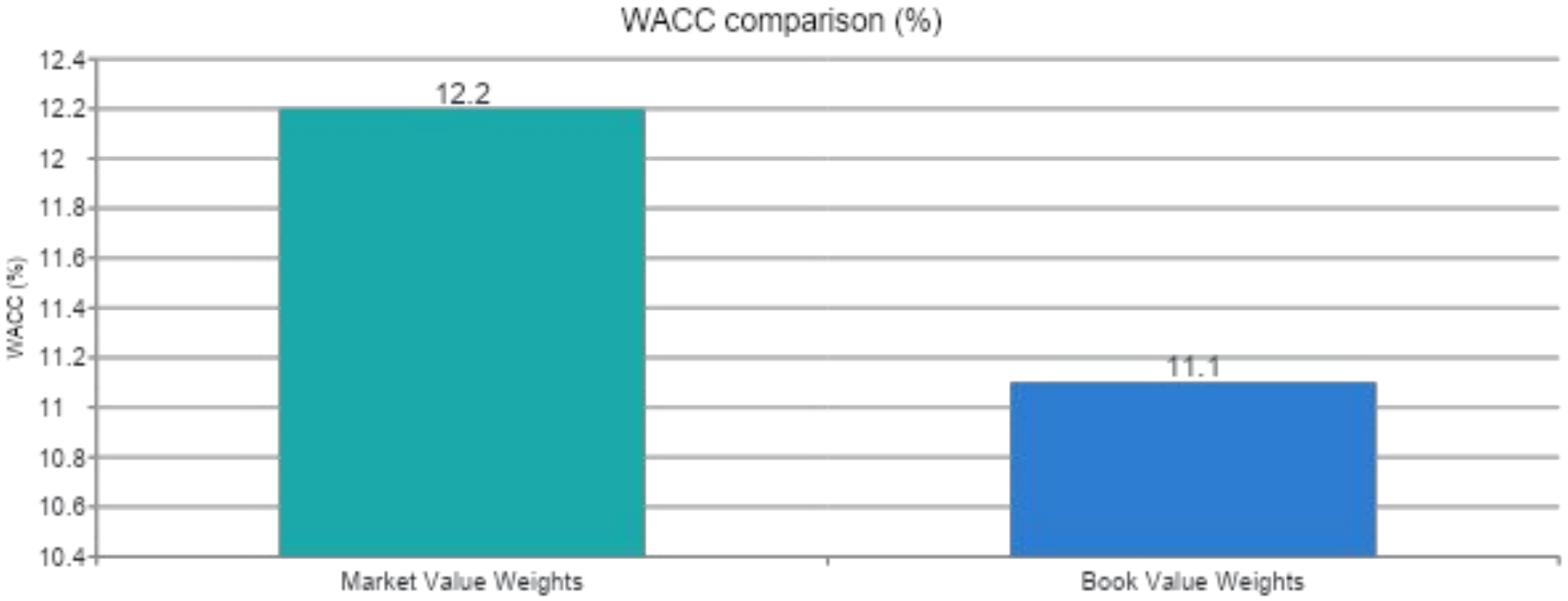
Reliance Industries — WACC Using Book Weights

Component	Cost	Weight	Weighted
Cost of equity: rE (unchanged)	13.3%	71.3%	9.48%
After-tax cost of debt: rD(1−TC) (unchanged)	5.6%	28.7%	1.61%

Book Value WACC = 11.1%

Weights: E/(E+D) = 8,20,000/11,50,000 = 71.3% D/(E+D) = 3,30,000/11,50,000 = 28.7% (₹ Crore)

Reliance Industries — Market vs Book WACC



Because Reliance's market value of equity (₹18,00,000 Cr) is far larger than its book equity (₹8,20,000 Cr), market weights place more weight on the (higher-cost) equity component — pushing market-value WACC above book-value WACC.

Company Snapshot — Infosys Ltd

NSE: INFY • IT Services — largely debt-free

Beta (β) **0.75**	Risk-free rate (Rf) **7.0%**	Market risk premium **6.0%**
Pre-tax cost of debt **6.5%**	Corporate tax rate **25%**	Market value — Equity (E) **₹6,20,000 Cr**
Market value — Debt (D) **₹8,000 Cr**	Book value — Equity (E) **₹90,000 Cr**	Book value — Debt (D) **₹8,000 Cr**

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Infosys — WACC Using Market Weights

Component	Cost	Weight	Weighted
Cost of equity: rE = 7.0% + 0.75 × 6.0%	11.5%	98.7%	11.35%
After-tax cost of debt: rD(1−TC) = 6.5% × 0.75	4.9%	1.3%	0.06%

Market Value WACC = 11.4%

Weights: E/(E+D) = 6,20,000/6,28,000 = 98.7% D/(E+D) = 8,000/6,28,000 = 1.3% (₹ Crore)

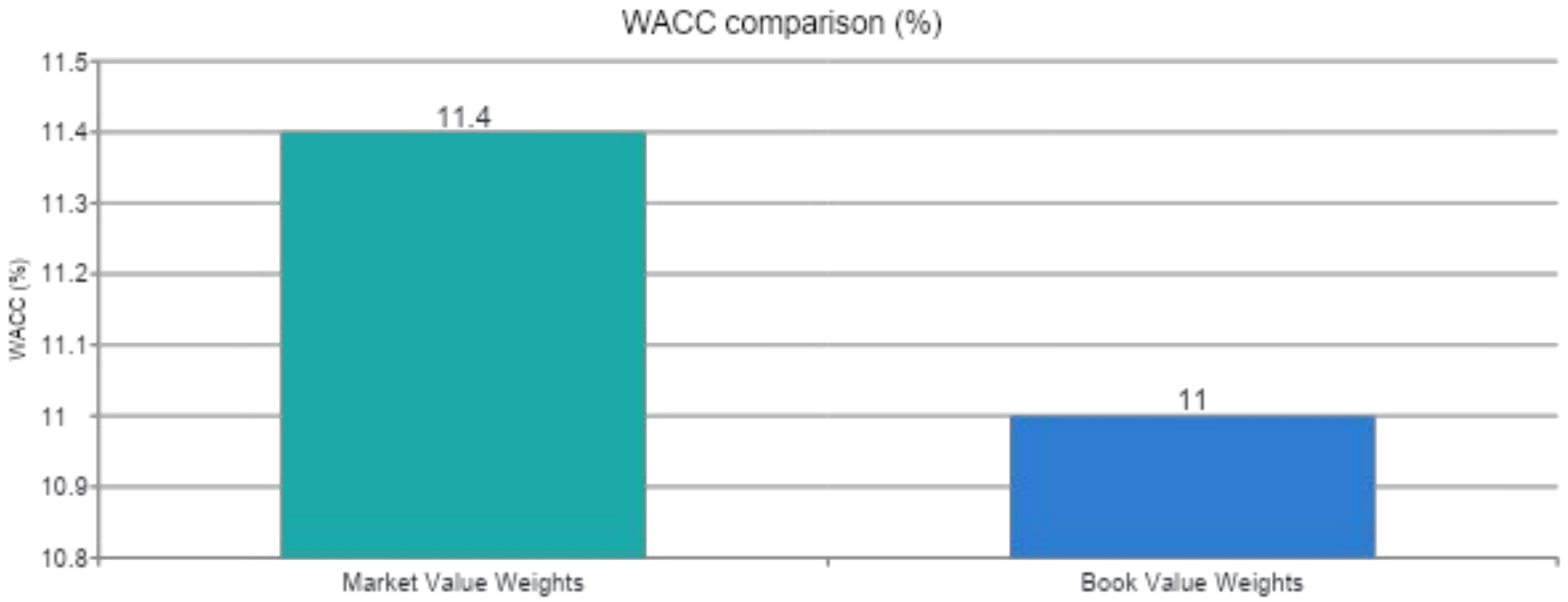
Infosys — WACC Using Book Weights

Component	Cost	Weight	Weighted
Cost of equity: rE (unchanged)	11.5%	91.8%	10.56%
After-tax cost of debt: rD(1−TC) (unchanged)	4.9%	8.2%	0.40%

Book Value WACC = 11.0%

Weights: E/(E+D) = 90,000/98,000 = 91.8% D/(E+D) = 8,000/98,000 = 8.2% (₹ Crore)

Infosys — Market vs Book WACC



Infosys carries very little debt either way, so the market-vs-book gap in WACC is small (11.4% vs 11.0%). Even so, book equity (₹90,000 Cr) understates the firm's true equity value (₹6,20,000 Cr) — a common feature of asset-light, high-growth businesses.

Summary

01

WACC is the blended rate of return a firm must earn to satisfy all of its capital providers.

02

It weights the after-tax cost of debt and the cost of equity (and preferred equity, if any) by their share of total capital.

03

Weights should ideally reflect the firm's target capital structure, measured in market value terms.

04

Book values are simpler to obtain but can materially understate equity value for high-growth or asset-light firms.

05

For Reliance, market-value WACC (12.2%) exceeded book-value WACC (11.1%) because market equity far exceeds book equity.

06

For Infosys, the gap was smaller (11.4% vs 11.0%) since the company carries very little debt under either method.

Thank You

Cost of Capital — Weighted Average Cost of Capital (WACC)

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